

```

clc; clear; close all;
rb_bw = 180e3;
users = 1:5;
alloc_RB = [10 12 8 15 5];
mod_eff = [2 4 6 4 2];
throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
disp('User Throughput (Mbps):')

```

User Throughput (Mbps):

```
disp(throughput')
```

```

3.6000
8.6400
8.6400
10.8000
1.8000

```

```

figure
subplot(2,1,1)
plot(users, throughput, '-o', 'LineWidth', 2)
title('OFDMA User Throughput')
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
subplot(2,1,2)
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
title('Resource Blocks vs Throughput')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on

```

